

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

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Abstract—Vision-language-action (VLA) policies map visual observations and language instructions to robot actions but can drift from successful behavior during closed-loop execution, raising the risk of task failure. We focus on recoverable execution deviations—where the rollout has left the demonstrated support yet retrying remains feasible—rather than irreversible failures requiring replanning. Existing solutions decouple detection and recovery into separate modules, often requiring human takeover data. We introduce DUET, a backbone-agnostic dual-action-head module that predicts relative actions for smooth nominal control and absolute waypoints for trajectory re-entry. Disagreement between the two action spaces signals rising failure risk, while the absolute waypoint supplies the retry direction. Asymmetric training keeps the relative branch sensitive to deviations and the absolute branch robust to visual disturbances. The module interfaces with shared vision-language features and can be attached to different VLA backbones. On LIBERO, DUET improves nominal success from 93.8% to 96.9% and action-perturbation success from 27.9% to 64.6% over its relative-only head. A controlled Franka protocol on a separately trained π_0 backbone is specified to test physical retry; its results remain pending.

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions to robot actions [1], [2], [3]. Trained exclusively on successful demonstrations, they can drift during closed-loop execution until the rollout can no longer reliably complete the task—an *execution failure*. Before that outcome becomes irreversible, the robot may already have departed from the observations, configurations, or action consequences represented by successful demonstrations. We call this intermediate condition an *execution deviation*: the rollout has left the support of demonstrated success, raising failure risk. Deviation and failure are related but not equivalent—an unseen state may still be handled successfully, while failures can also arise from compounding errors within seemingly familiar observations. Our focus is the practical regime in which a deviation has elevated failure risk but the task can still be completed by returning toward demonstrated behavior.

Existing approaches decouple detection and recovery: an external monitor flags the failure [4], and a separate controller, often trained on human takeover data, executes a correction [5], [6], [7]. This design is costly and loosely coupled—a detector only signals that execution has become abnormal; it does not specify how to return to behavior that can still complete the task. We ask whether a VLA can use

its own action predictions to detect a recoverable execution deviation and generate a retry direction from successful demonstrations alone, without failure labels or takeover data.

DUET is motivated by a complementary asymmetry between two action representations. Relative-action policies produce smooth motion around the demonstrated trajectory, but once a grasp or pose deviates, a local command provides no target for returning. Absolute-action policies are less suited to fine local control, yet their waypoints can point back toward a reachable portion of the demonstrated trajectory. DUET learns the same behavior in both spaces: a relative head for nominal execution and an absolute head for trajectory re-entry. After transporting the absolute waypoint into the live frame, we compare it with the relative action. Their disagreement measures whether local control remains consistent with the global reference and serves as an endogenous signal of rising failure risk. It is not a universal OOD score; it is a task-specific inconsistency signal anchored to successful behavior.

The dual-head module interfaces only with the backbone’s shared vision-language representation, requiring no modification to the backbone architecture or pretraining. This makes DUET a transferable add-on: we train it on StarVLA [3] for simulation and on π_0 (OpenPi) [8] for real-robot deployment without architecture-specific adaptation.

At inference time, a soft gate keeps relative control in the nominal regime and shifts toward the waypoint-induced correction when disagreement indicates a recoverable deviation; control returns to relative mode after the two predictions realign. DUET is a failure-aware retry mechanism, not a general recovery planner: it does not address irreversible failures, hardware faults, or cases requiring a new task strategy.

Our contributions are threefold:

- We formulate failure-aware retry as detecting recoverable execution deviations from successful behavior and re-entering its support. Cross-space disagreement between relative and absolute action heads serves as an endogenous failure-risk signal requiring no failure labels, outperforming dedicated detectors on TPR and AUROC (§IV-B).
- We propose DUET, a dual-action-head VLA in which the same disagreement that detects deviations simultaneously provides the waypoint-based recovery direction, unifying detection and correction in a single mechanism while preserving smooth nominal control (§IV-C, §IV-D).
- We evaluate DUET on StarVLA in LIBERO simulation

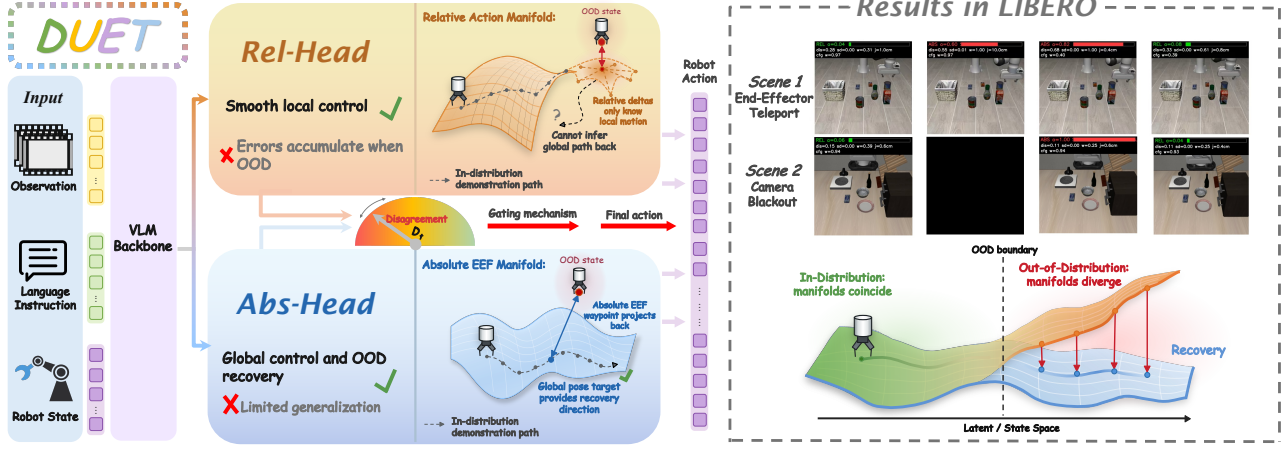


Fig. 1. Failure-aware retry with dual action spaces. Relative and absolute predictions agree during nominal execution but diverge when a recoverable deviation drives the rollout OOD, exposing an endogenous disagreement score D_t . Relative actions provide precise local motion, whereas absolute waypoints bring the robot back toward a demonstrated execution path for retry.

(§IV-D) and define a matched clean/disturbed protocol on a separately trained π_0 (OpenPi) controller across four Franka tasks (§IV-F) to test the design’s applicability across backbones.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-Dagger uses diffusion uncertainty to request intervention [9]; Sentinel monitors action inconsistency and task progress [10]. FAIL-Detect formulates failure detection as sequential OOD testing on successful executions [4]; FIPER combines representation novelty with action uncertainty [11].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful and failed rollouts [12]; AHA trains a VLM to detect and diagnose failures [13]. All of these methods—including FAIL-Detect’s logpZO [4]—produce a scalar alarm but no recovery action. DUET instead detects execution deviations through disagreement between two action spaces; the same inconsistency that signals rising failure risk provides the waypoint-based recovery direction, coupling detection and correction without additional supervision.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAGger queries experts on policy-induced states [5]; intervention-based methods collect human takeovers [14], [6]; IntervenGen transforms limited interventions into corrective trajectories [7]. These reduce compounding errors but require additional recovery data.

Foundation-model systems use semantic supervisors. RACER learns from language-annotated recovery trajectories [15]; AHA and VLM controllers diagnose failures and guide correction [13], [16]. In all cases detection and recovery remain separately trained or prompted. DUET couples both

without extra data: absolute states from successful demonstrations serve as recovery anchors, and cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Dual Action Spaces and Architecture

Given images o_t , instruction l , and proprioceptive state s_t , DUET predicts relative action and absolute waypoint chunks over a horizon K :

$$\begin{aligned}\hat{\Delta}A_t &= f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \\ \hat{W}_t &= f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}.\end{aligned}\quad (1)$$

Commands decompose as $a_t = [a_{p,t}, a_{g,t}]$, with six pose dimensions $a_{p,t}$ and one gripper dimension $a_{g,t}$. Relative outputs are operational-space-control deltas; absolute outputs encode future end-effector poses and gripper states. Only pose commands are fused; deployed gripper control comes from the relative head.

We model successful end-effector trajectories as a manifold $\mathcal{M}_l \subset SE(3)$. Near this support, relative commands describe local tangent motion in $T_{p_t}\mathcal{M}_l$, while absolute waypoints reference global coordinates on $\mathcal{M}_l$. After a deviation, a relative command continues from the displaced pose, whereas an absolute waypoint supplies a persistent target toward demonstrated behavior. Their transported predictions should align in-domain; OOD execution can expose inconsistency between local continuation and the global reference. This geometric interpretation motivates detection without guaranteeing it.

Both heads share a VLM encoder (Fig. 2). The relative head attends to its full hidden sequence and receives s_t through a state encoder. The absolute head excludes explicit proprioception and state tokens from conditioning, preserving an observation-conditioned reference. State tokens, when present, follow visual-language tokens in the causal encoder and are omitted from absolute-branch pooling.

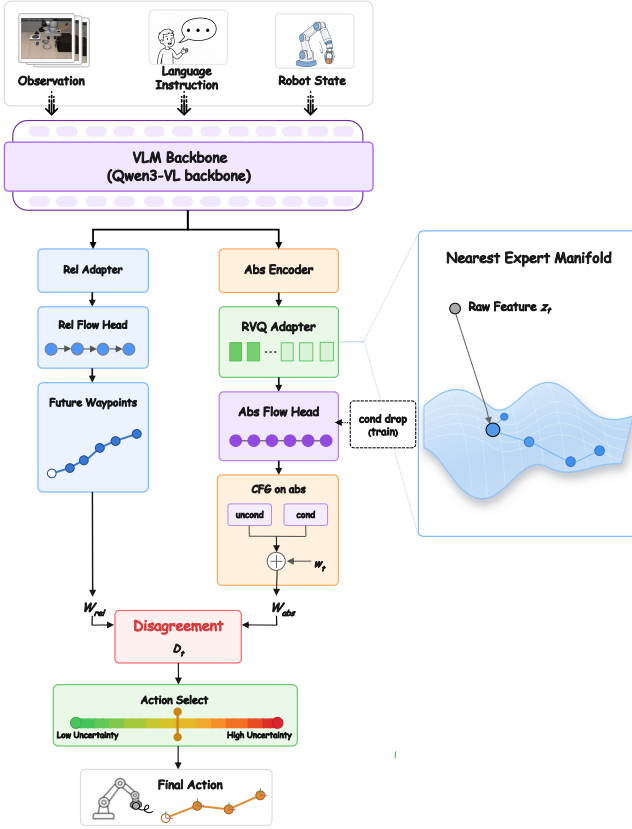


Fig. 2. DUET combines a shared encoder with state-aware relative and RVQ-conditioned absolute heads. Their disagreement guides OOD detection and closed-loop waypoint retry.

B. Disagreement-Preserving Learning

Successful demonstrations supply paired delta actions and normalized future states; no failure labels or human takeover actions are required. We optimize a weighted sum of the two flow-matching losses [8] and an RVQ commitment loss, without cross-head consistency supervision that could suppress detection evidence. RVQ quantizes pooled state-free visual-language features into a discrete condition atlas learned from demonstrations, anchoring absolute predictions [17], [18]; its residual is retained only as auxiliary telemetry.

Asymmetric augmentation trains the absolute head and RVQ on clean and perturbed observations, but computes relative-head loss only on clean samples, encouraging a robust reference and deviation-sensitive local control. For classifier-free guidance (CFG) [19], condition dropout replaces absolute-head conditions with learned null tokens to learn conditional and marginal waypoint flows.

C. Endogenous OOD Detection and Retry

Both branches are unnormalized before comparison in the tangent coordinates of the live pose. For live pose $p_t = (x_t, R_t)$ and predicted waypoint $\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$, the waypoint-induced retry direction is

$$c_t = \text{clip} \left(\left[\frac{(\hat{x}_t - x_t)/\sigma_p}{\text{Log}(\hat{R}_t R_t^{-1})/\sigma_R} \right], -1, 1 \right), \quad (2)$$

where Log denotes the three-vector rotation logarithm and σ_p, σ_R convert pose error into action units. Comparing it with the relative pose command $\hat{\delta}a_{p,t}$ gives endogenous OOD evidence:

$$D_t = \frac{\|c_t - \text{clip}(\hat{\delta}a_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (3)$$

Here D_t supplies scalar detection evidence, while c_t supplies a six-dimensional retry direction. The auxiliary S_t averages coordinatewise distances outside the training pose bounds, normalized by each coordinate's training range. The control score q_t and fusion weight are

$$q_t = D_t + \lambda S_t, \quad (4)$$

$$\alpha_t = \sigma(k(q_t - \tau)).$$

Here σ is the sigmoid, k its slope, and τ the switching threshold; λ weights state distance. The executed command is

$$a_{p,t} = (1 - \alpha_t)\hat{\delta}a_{p,t} + \alpha_t c_t, \quad (5)$$

$$a_{g,t} = \hat{\delta}a_{g,t}.$$

CFG strength follows $w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}$ for the next chunk. When the original trajectory remains feasible, rising disagreement shifts control toward c_t for retry, and realignment restores relative execution.

IV. EXPERIMENTS

We organize the experiments around three questions:

- (Q1) **Endogenous OOD detection:** Can DUET identify execution states that are inconsistent with successful behavior using only successful demonstrations?
- (Q2) **Nominal execution:** Does the dual-action mechanism preserve normal task execution?
- (Q3) **Failure-aware retry:** When an OOD deviation remains recoverable, can DUET use its detected inconsistency to re-enter successful behavior and complete the task?

We evaluate these questions with matched LIBERO experiments, controlled execution disturbances, detector-score analysis, component ablations, and a real-robot protocol. Detector quality, nominal task success, and closed-loop retry success are reported separately.

A. Evaluation Protocol

We evaluate on four LIBERO suites [21]—Spatial, Object, Goal, and LIBERO-10—with 10 tasks and 50 initial states per task (500 episodes per suite). Three methods are compared:

- *StarVLA* [3]: relative-action-only VLA trained with the same data and budget but without the absolute head (external baseline).
- *DUET-Rel*: relative head of the dual-action checkpoint with $\alpha_t=0$ (isolates co-training from fusion).
- *DUET*: full adaptive controller (Eq. (5)).

StarVLA $\rightarrow$ DUET-Rel measures co-training; DUET-Rel $\rightarrow$ DUET isolates the retry mechanism. Architecture, data, task, initial state, and perturbation are held fixed within each

TABLE I
EXECUTION-FAILURE PROTOCOLS AND THEIR CORRESPONDENCE TO THE INTRODUCTION.

Family	Protocol	What becomes OOD	Introduction example
Action execution	Action perturbation	Induced trajectory / command-to-motion map	Execution drift after locally plausible commands
Robot displacement	EEF teleport	Robot-arm pose	Unexpected displacement of the robot arm
Object / grasp	Vertical grasp drop	Robot-object configuration	Dropped object or failed hold
Visual occlusion	Camera blackout	Current visual observation	Transient occlusion
Contact / scene	External force	Robot-world interaction	External disturbance during contact

TABLE II
FAILURE-DETECTION COMPARISON UNDER MATCHED PROTOCOLS. TPR, TNR, BALANCED ACCURACY (BAL.-ACC.), AUROC, AND LOC.@16 ARE HIGHER-IS-BETTER; DETECTION DELAY IS LOWER-IS-BETTER. BOLD VALUES INDICATE THE BEST RESULT IN EACH COLUMN.

Method	TPR $\uparrow$	TNR $\uparrow$	Bal.-Acc. $\uparrow$	AUROC $\uparrow$	Delay (steps) $\downarrow$	Delay (s) $\downarrow$	Loc.@16 $\uparrow$
DUET	0.900	0.265	0.582	0.845	28.3	1.41	0.523
logpZO [4]	0.889	0.243	0.566	0.817	22.7	1.14	0.569
RND [20], [4]	0.842	0.361	0.602	0.549	26.1	1.31	0.330
PCA-kmeans [4]	0.884	0.283	0.583	0.625	20.3	1.01	0.503
STAC [10]	0.839	0.311	0.575	0.494	25.8	1.29	0.260

pair. Simulation uses StarVLA as the backbone; real-robot experiments (§IV-F) switch to π_0 (OpenPi) [8], testing cross-backbone transfer with identical module design and fusion logic.

Perturbations are scheduled at a fixed policy step with deterministic seeds. DUET and DUET-Rel are paired by identical (task, initial state) keys; we report closed-loop task success with McNemar p -values. When a perturbation has a semantic precondition (e.g., grasp drop requires a held object), we report injection coverage and retain uninjected episodes. Table I maps each protocol to the failure types in the introduction; we do not pool heterogeneous disturbances into a single average.

B. (Q1) Endogenous OOD Detection

To answer **Q1**, Table II compares cross-space disagreement D_t with success-only detectors from FAIL-Detect [4] and STAC [10] under matched protocols.

DUET improves detection discrimination while keeping detection delay close to prior methods: 1.41 s versus 1.01–1.31 s. It achieves the highest AUROC (0.845), exceeding the strongest baseline, logpZO (0.817), and substantially outperforming RND, PCA-kmeans, and STAC (0.494–0.625). DUET also attains the highest TPR (0.900), although its balanced accuracy is not the best. These results show that disagreement between the two action spaces identifies execution deviations without failure labels, with improved ranking quality at a modest additional delay.

C. (Q2) Nominal Execution

Q2 asks whether failure-aware retry preserves ordinary execution when no disturbance is injected. DUET attains 96.9% average success (Table III), matching the reported $\pi_{0.5}$ result. External rows serve as benchmark context only; the controlled comparison is DUET versus DUET-Rel on the same checkpoint. Adaptive fusion raises the average

TABLE III
NOMINAL TASK SUCCESS (%) ON LIBERO. EXTERNAL ROWS ARE REPORTED RESULTS AND ARE INCLUDED ONLY FOR BENCHMARK CONTEXT. STARVLA USES JOINT 30K TRAINING; DUET USES SUITE-SPECIFIC 10K CHECKPOINTS.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET	98.8	99.4	97.6	91.8	96.9
DUET-Rel	98.0	99.2	92.0	86.0	93.8

from 93.8% to 96.9%, with the largest lifts on Goal (+5.6) and LIBERO-10 (+5.8). Even without injected perturbation, rollouts can drift from the demonstrated support; the absolute waypoint provides a return target that the relative-only controller lacks.

D. (Q3) Failure-Aware Retry under Execution Disturbances

Q3 asks whether DUET can complete a feasible task after an OOD execution deviation. Across the four primary disturbance protocols in Table IV, the answer is yes at the aggregate level. Compared with DUET-Rel, DUET raises success from 27.9% to 64.6% under action perturbation (+36.7 pp), from 34.5% to 59.4% after EEF teleport (+24.9 pp), from 89.2% to 95.2% after grasp drop (+6.0 pp; coverage $\geq 98.8\%$), and from 73.5% to 90.8% during camera blackout (+17.3 pp). DUET also exceeds StarVLA in all four protocols.

The largest gains occur under action perturbation and EEF teleport, where displaced relative control lacks a return target; the smaller grasp-drop gain reflects the strong relative-only baseline. The improvement is not universal at the cell level: DUET is 1.8 pp below DUET-Rel for medium camera blackout on Object, and the Spatial teleport gains are not statistically significant. Figure 3 shows the

TABLE IV

CLOSED-LOOP SUCCESS (%) UNDER FOUR DISTURBANCE FAMILIES ($N=500$ PER CELL). OVERALL ROWS AGGREGATE MEDIUM AND SEVERE RESULTS ACROSS ALL FOUR SUITES. PER-SUITE REL $\rightarrow$ DUET Δ HAS MCNEMAR $p<10^{-3}$ EXCEPT TELEPORT/SPATIAL (MEDIUM $p=0.21$, SEVERE $p=0.067$) AND BLACKOUT/OBJECT MEDIUM.

Suite	Medium				Severe			
	StarVLA	Rel.	DUET	Δ	StarVLA	Rel.	DUET	Δ
<i>(i) Action perturbation (step 5)</i>								
Spatial	60.0	54.0	74.0	+20.0	14.0	12.8	43.2	+30.4
Object	41.6	48.8	84.4	+35.6	7.8	7.8	60.6	+52.8
Goal	52.2	47.2	74.2	+27.0	18.8	10.6	52.6	+42.0
LIBERO-10	36.2	36.6	78.6	+42.0	6.2	5.4	49.0	+43.6
Overall	29.6	27.9	64.6	+36.7	<i>med+sev combined</i>			
<i>(ii) EEf teleport (step 39)</i>								
Spatial	55.4	54.6	57.6	+3.0	46.8	48.2	52.6	+4.4
Object	43.4	36.2	76.0	+39.8	21.2	20.8	68.4	+47.6
Goal	41.6	33.4	52.4	+19.0	30.4	28.2	50.6	+22.4
LIBERO-10	40.4	35.8	60.8	+25.0	21.4	18.4	56.8	+38.4
Overall	37.6	34.5	59.4	+24.9	<i>med+sev combined</i>			
<i>(iii) Vertical grasp drop (Object only, medium, step 67)</i>								
Object	90.6	89.2	95.2	+6.0	—			
<i>(iv) Camera blackout (step 23)</i>								
Spatial	81.0	88.0	94.4	+6.4	42.4	63.2	95.2	+32.0
Object	91.8	98.6	96.8	−1.8	64.2	95.0	97.2	+2.2
Goal	65.4	86.0	86.4	+0.4	35.4	65.8	85.8	+20.0
LIBERO-10	60.0	65.6	84.2	+18.6	24.4	25.6	86.2	+60.6
Overall	58.1	73.5	90.8	+17.3	<i>med+sev combined</i>			

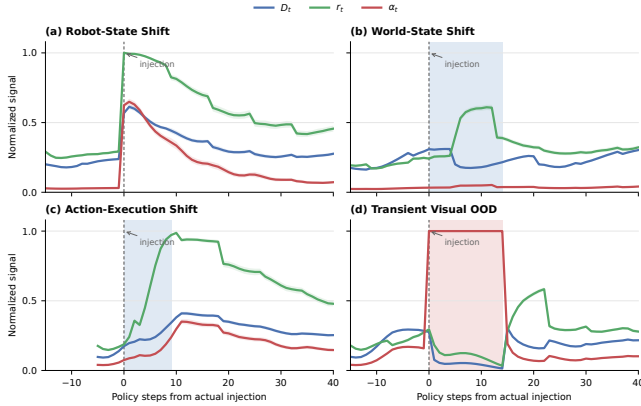


Fig. 3. Controller dynamics on the Object suite under (a) end-effector teleport, (b) vertical grasp drop, (c) action perturbation, and (d) camera blackout. Traces are aligned to the actual injection time and show means with pointwise bootstrap 95% confidence intervals over successfully injected episodes. Shading denotes the perturbation or forced-hold window.

retry mechanism: after a disturbance, disagreement raises α_t and shifts pose control toward the absolute waypoint; after re-entry, α_t falls and relative control resumes. In a supplementary contact-force check, DUET improves success from 92.22% to 95.95% (+3.73 pp), indicating that adaptive fusion remains stable in contact-rich tasks. Together, these results show that cross-space inconsistency can both trigger and guide a successful retry, directly answering **Q3**.

E. Ablations

Table V separates inference-time controls from training-time replacements. Disabling the absolute branch (DUET-

τ - k sensitivity heatmap

Left: clean success (%); right: action-perturbation medium success (%).
Horizontal axis: k ; vertical axis: τ . Color encodes success rate.

TODO: generate with
`matplotlib.pyplot.imshow` or
`seaborn.heatmap`.

Fig. 4. Sensitivity of clean and OOD success (%) to the gate parameters τ and k on the Object suite. The dashed rectangle highlights the operating region where both metrics exceed their respective thresholds.

Rel) accounts for all reported gains; fixing CFG at $w=1$ leaves performance unchanged, so the benefit comes from the waypoint itself. Absolute-only ($\alpha_t=1$) and fixed-fusion ($\alpha_t=0.5$) controls collapse performance, confirming the disagreement gate is essential. Training ablations (RVQ $\rightarrow$ MLP, symmetric augmentation, state-aware absolute head) are preregistered as TODO.

a) *Fusion weight on clean trajectories.*: Across 2,000 clean episodes, the mean α_t is TODO, the 95th percentile is TODO, and only TODO% of steps exceed 0.5—confirming that the fusion weight stays near zero during nominal execution.

b) *Sensitivity to τ and k .*: Figure 4 sweeps τ and k on the Object suite. A broad plateau with clean $> 90\%$ and OOD $> 80\%$ indicates low sensitivity; we select τ =TODO, k =TODO from this region.

TABLE V

ABLATION SUCCESS (%). FULL DUET AND DUET-REL AVERAGE FOUR SUITES ($N=2000$ FOR CLEAN AND MEDIUM TELEPORT; $N=4000$ FOR MEDIUM+SEVERE ACTION PERTURBATION). $\dagger$: OBJECT-ONLY, WITH MEDIUM DISTURBANCES ($N=500$ PER PROTOCOL); MATCHED DUET SCORES ARE 99.2/76.0/84.4% IN COLUMN ORDER.

Type	Variant	Checkpoint	Clean	Robot-state	Action-execution	Isolates
Inference	Full DUET	shared	96.9	61.7	64.6	Complete controller
	DUET-Rel ($\alpha = 0$)	shared	93.8	40.0	27.9	Absolute retry branch
	Absolute-only ($\alpha = 1$) †	shared	5.2	1.1	0.8	Cost of losing local control
	Fixed fusion ($\alpha = 0.5$) †	shared	46.2	39.4	41.0	Adaptive switching
	No adaptive CFG †	shared	88.2	52.6	57.2	CFG scale schedule
Training	RVQ $\rightarrow$ MLP	retrained	TODO	TODO	TODO	Discrete waypoint anchoring
	Symmetric augmentation	retrained	TODO	TODO	TODO	Complementary specialization
	State-aware ABS	retrained	TODO	TODO	TODO	State-free global reference

Real-robot task setups

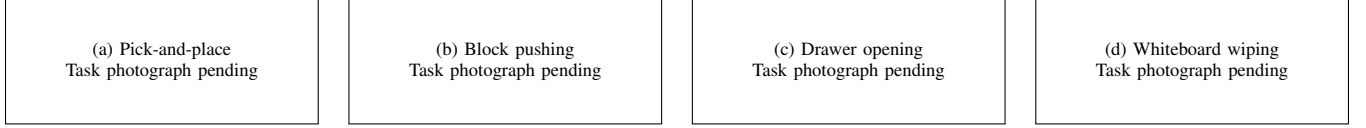


Fig. 5. Four Franka task configurations. Each panel will show the target, disturbance stage, and measured displacement direction.

TABLE VI

FRANKA SUCCESS RATE (%), 20 TRIALS PER TASK, METHOD, AND CONDITION. D1/D2 DENOTE THE TWO DISTURBANCE TYPES. DASHES INDICATE UNMEASURED RESULTS. ALL METHODS USE π_0 .

Method	Clean	D1	D2
<i>Pick-and-place</i>			
π_0	—	—	—
DUET-Rel	—	—	—
DUET	—	—	—
<i>Block pushing</i>			
π_0	—	—	—
DUET-Rel	—	—	—
DUET	—	—	—
<i>Drawer opening</i>			
π_0	—	—	—
DUET-Rel	—	—	—
DUET	—	—	—
<i>Whiteboard wiping</i>			
π_0	—	—	—
DUET-Rel	—	—	—
DUET	—	—	—
<i>Equal-weight mean</i>			
π_0	—	—	—
DUET-Rel	—	—	—
DUET	—	—	—

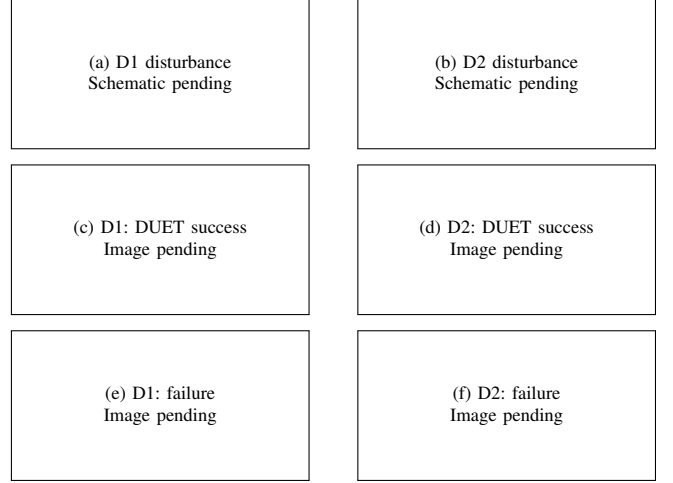


Fig. 6. Two disturbance types (top), successful responses (middle), and failed responses (bottom). Columns correspond to D1 and D2; photographs will identify the method and actual outcome.

F. Real-Robot Evaluation

We evaluate the same dual-head design on π_0 (OpenPi) [8] and a Franka Emika Panda. The experiment tests whether the simulated retry mechanism remains useful under physical disturbances; it does not assume zero-shot transfer between separately trained backbones.

a) Platform and protocol.: We use a wrist-mounted RealSense camera and successful demonstrations only. Table VI covers four tasks, three methods, three conditions, and 20 trials per cell (720 trials): π_0 , π_0 +DUET-Rel with

fusion disabled, and full π_0 +DUET. The latter two share a checkpoint. Data splits, inputs, training budgets, execution rates, and time limits are matched; necessary differences are documented. Trials use matched scenes, randomized order, and complete resets.

b) Tasks and disturbances.: Each task is evaluated without disturbance and under two disturbance types, denoted D1 and D2 until their physical protocols are finalized. Figure 6 reserves one schematic per type and successful/failed response examples. Trigger stage and disturbance

magnitude are fixed before formal testing using separate pilot trials; the original task must remain feasible. The same protocol is applied across methods, with no human correction. Trials that fail before injection remain in the success-rate denominator.

Success must persist for two seconds: the released or pushed object lies inside its fixed target region, the drawer reaches its calibrated opening, or wiping removes at least 90% of the predefined stain area while retaining the tool. Existing validated wiping criteria take precedence and are frozen before testing. Success rate is the sole evaluation metric, computed from 20 trials per task, method, and condition; the mean weights all four tasks equally.

c) *Qualitative outcomes.*: Recorded images illustrate the two disturbance types and their successful and failed responses. Each example is linked to a trial identifier, method, and condition. Final task completion determines success; selected images explain the behavior and do not replace the aggregate success rates.

Comparisons under D1 and D2 test whether DUET improves task completion after different disturbances. The clean condition tests nominal success. The shared-checkpoint comparison between DUET-Rel and DUET isolates the contribution of online fusion.

G. Discussion

The simulation results establish the detector and fusion mechanism; the Franka protocol tests whether the same logic improves success under two physical disturbance types while preserving clean success. Qualitative images will illustrate successful and failed responses. The real-robot results remain pending.

V. LIMITATIONS

DUET assumes absolute state labels aligned with relative action chunks—natural in LIBERO and many robot logs but requiring accurate calibration on the Franka platform. The absolute waypoint is a demonstration anchor, not an object tracker: a world-state shift can invalidate the reference even when the robot remains nearby. On LIBERO, S_t is uninformative, so gating relies on the residual to the predicted waypoint and cannot distinguish “the arm left the target” from “the target moved.” Complete visual dropout requires a validity guard and cached waypoint rather than disagreement-based detection. Retry is limited to recoverable deviations; irreversible failures or tasks requiring a new strategy fall outside the mechanism.

VI. CONCLUSION

We presented DUET, a dual-head module for endogenous deviation detection and trajectory re-entry. Relative actions support local control, while stateless absolute waypoints provide a retry target when execution leaves demonstrated behavior. Simulation results establish the mechanism; the controlled Franka protocol defines how to test its physical retry benefit and nominal execution cost on a separately

trained π_0 backbone. The scope remains limited to recoverable deviations; stale references, irreversible failures, and fully invalid observations require additional mechanisms.

ACKNOWLEDGMENT

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